

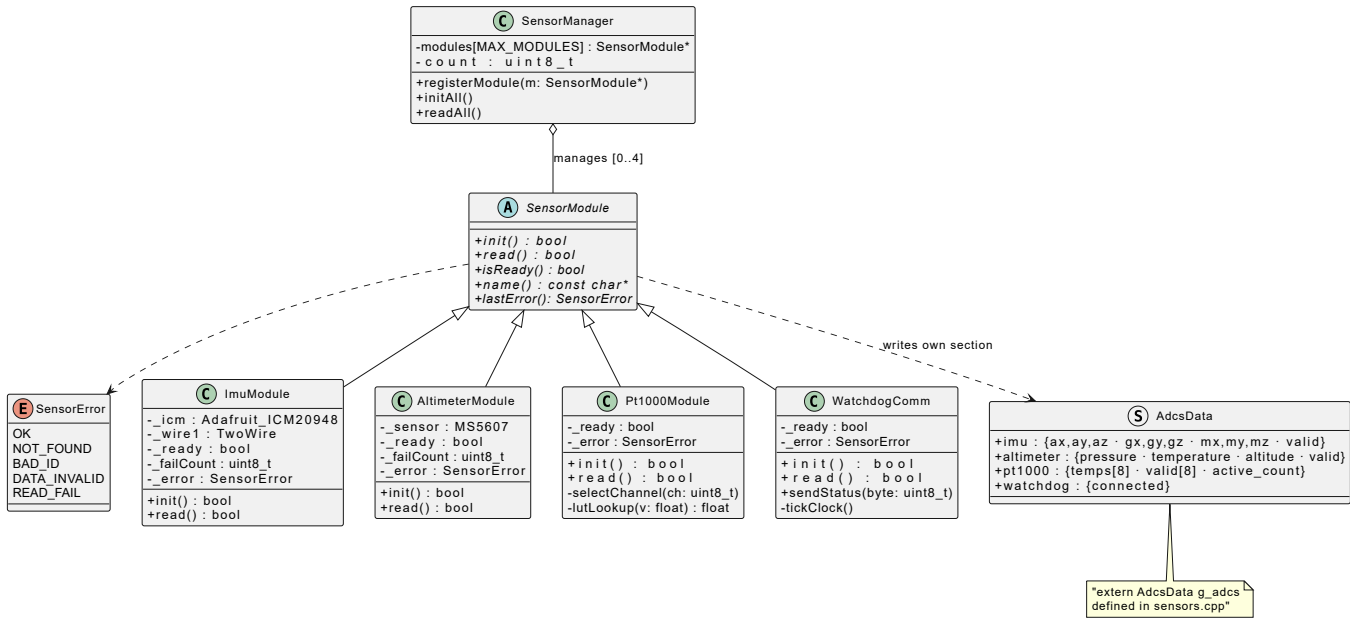
Firmware Architecture — SAFARI ADCS V1.2

Platform: Raspberry Pi Pico (RP2040) · Arduino-Mbed · PlatformIO
Revision: V2.1 — 2026-06-10

1. Overview

The firmware is organized around a central **SensorManager** that orchestrates up to four independent sensor modules. Each module implements the same abstract interface (**SensorModule**). Results are written to a single shared data bus (**AdcsData g_adcs**). No module depends on another.

2. Class Hierarchy

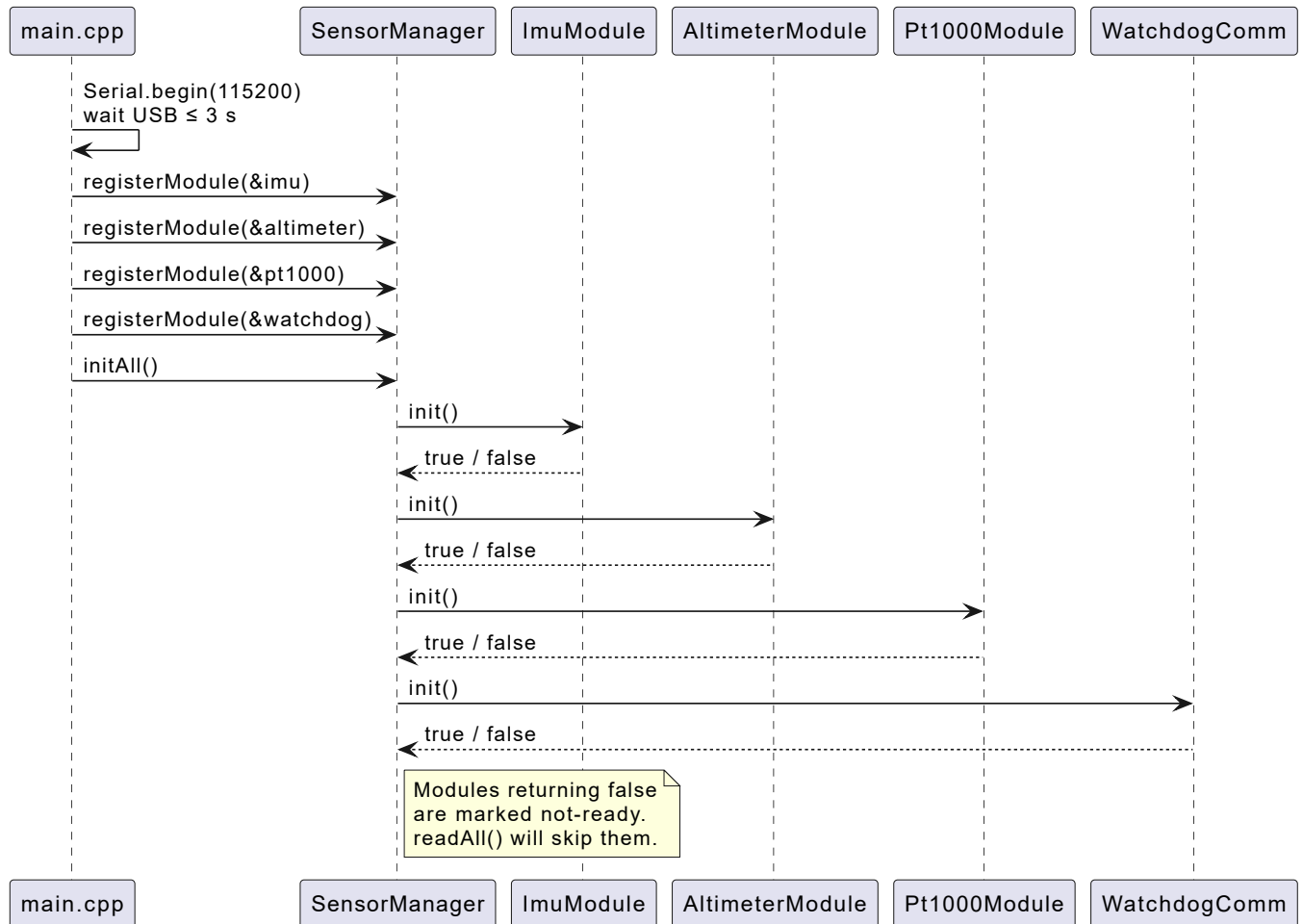


3. Shared Data Bus — AdcsData

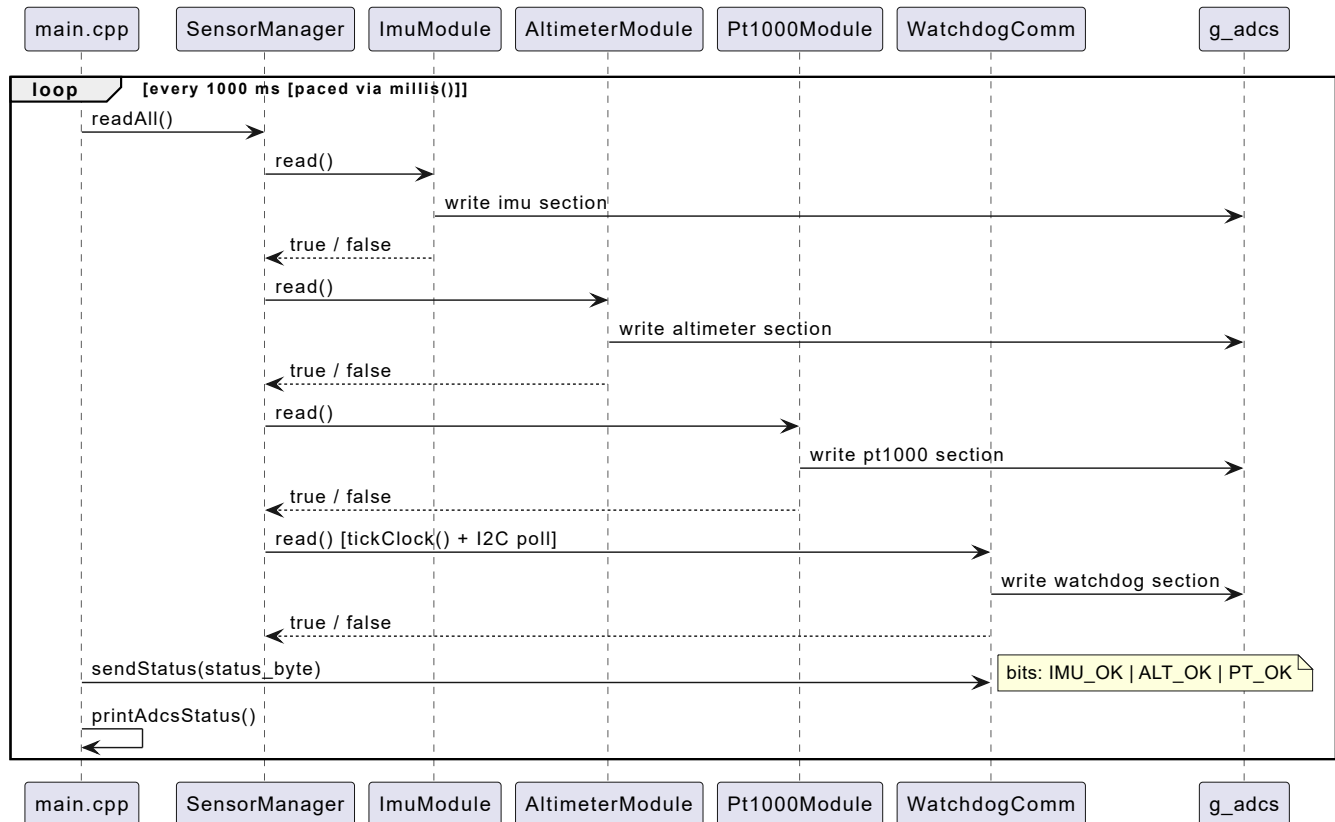
g_adcs : AdcsData
imu.ax / ay / az : float ' m/s ² imu.gx / gy / gz : float ' rad/s imu.mx / my / mz : float ' μT imu.valid : bool
altimeter.pressure : float ' hPa altimeter.temperature: float ' °C altimeter.altitude : float ' m altimeter.valid : bool
pt1000.temps[8] : float ' °C pt1000.valid[8] : bool pt1000.active_count : uint8_t
watchdog.connected : bool

Each module writes **only its own section**. `main.cpp` reads all sections after `readAll()`.

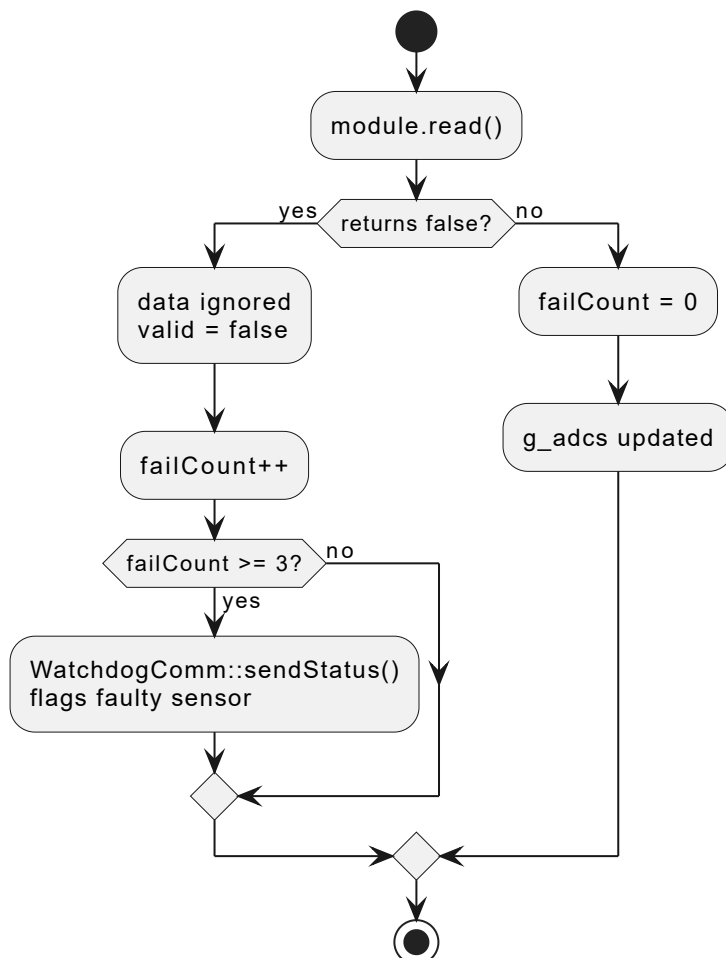
4. Initialization Sequence — `setup()`



5. Main Loop Sequence — `loop()` (period = 1000 ms)



6. Recovery Policy



Event	Immediate reaction	After 3 failures
<code>init()</code> → false	module disabled, Serial log	stays disabled
<code>read()</code> → false	<code>valid = false</code>	watchdog notified via <code>sendStatus()</code>
Locked I2C bus	<code>Wire.end()</code> / <code>Wire.begin()</code>	module permanently disabled